Evaluation runs: 2pc-ds (two-pc-disel)

generated 2026-11-03 from brain-data b2ae60f86 2026-11-03 17:57:09 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 9a0629e, merge join 367cdc6, JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{spec}, G_{impl} = the graphs of the specification and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G^{expl}_{spec}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

stateless

stateless: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
2	400	256	conforms	0.2	2	1 h, 250 GB	16	2384	brain03, 2026-11-02
3	104976	82944	conforms	212.6	4	1 h, 250 GB	64	2574748	brain03, 2026-11-02
4	95883264		🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-02

stateless: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
2	400	256	conforms	0.2	2	1 h, 250 GB	16	2370	brain03, 2026-11-02
3	104976	82944	conforms	208.7	2	1 h, 250 GB	64	2559952	brain03, 2026-11-02
4	95883264		🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-02

stateless: pair **skip-ack-loop**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
2	400	64	64	violation	1.3	2	1 h, 250 GB	64	25600	brain03, 2026-11-02
3	104976			🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-02

stateless: pair **skip-ack-loop**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
2	400	64	64	violation	1.3	4	1 h, 250 GB	64	25600	brain03, 2026-11-02
3	104976			🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-02

Not run (stateless). skipped: a smaller size had reached its limit: 4 skip-ack-loop fifo, 4 skip-ack-loop bag, 5 skip-ack-loop bag, 5 skip-ack-loop fifo, 5 correct bag, 5 correct fifo, 6 skip-ack-loop bag, 6 correct bag, 6 correct fifo, 6 skip-ack-loop fifo, 7 skip-ack-loop bag, 7 correct bag, 7 skip-ack-loop fifo, 7 correct fifo, 8 correct fifo, 8 correct bag, 8 skip-ack-loop bag, 8 skip-ack-loop fifo, 9 correct bag, 9 correct fifo, 9 skip-ack-loop fifo, 9 skip-ack-loop bag.

planned but never scheduled (grid points beyond the ladder): 4 bag skip-ack-loop, 4 bag correct, 5 bag correct, 5 bag skip-ack-loop.

stateful

stateful: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2	400	256	conforms	0.1	4	1 h, 250 GB	brain02, 2026-11-02
3	104976	82944	conforms	6.8	2	1 h, 250 GB	brain02, 2026-11-02
4	95883264	84934656	conforms	2212.4	236	48 h, 250 GB	brain05, 2026-11-02
5			(1 h)			1 h, 250 GB	brain02, 2026-11-02

stateful: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2	400	256	conforms	0.1	4	1 h, 250 GB	brain02, 2026-11-02
3	104976	82944	conforms	7.7	4	1 h, 250 GB	brain02, 2026-11-02
4	95883264	84934656	conforms	2209.8	240	1 h, 250 GB	brain02, 2026-11-02
5			(1 h)			1 h, 250 GB	brain02, 2026-11-02

stateful: pair **skip-ack-loop**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
2	400	64	64	violation	0.1	4	1 h, 250 GB	brain02, 2026-11-02
3	104976	2304	2304	violation	3.9	4	1 h, 250 GB	brain02, 2026-11-02
4	95883264	147456	147456	violation	1089.8	240	48 h, 250 GB	brain05, 2026-11-02
5				(1 h)			1 h, 250 GB	brain02, 2026-11-02

stateful: pair **skip-ack-loop**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
2	400	64	64	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
3	104976	2304	2304	violation	3.8	4	1 h, 250 GB	brain02, 2026-11-02
4	95883264	147456	147456	violation	1115.0	236	1 h, 250 GB	brain02, 2026-11-02
5				(1 h)			1 h, 250 GB	brain02, 2026-11-02

Not run (stateful). planned but never scheduled (grid points beyond the ladder): 5 correct bag, 5 bag correct, 5 bag skip-ack-loop, 5 skip-ack-loop bag, 6 bag skip-ack-loop, 6 bag correct, 7 bag correct, 7 bag skip-ack-loop.

left out by the table rules (ok): 4 correct bag, 4 skip-ack-loop bag.

skipped: a smaller size had reached its limit: 6 correct bag, 6 correct fifo, 6 skip-ack-loop fifo, 6 skip-ack-loop bag, 7 skip-ack-loop fifo, 7 correct bag, 7 skip-ack-loop bag, 7 correct fifo, 8 correct bag, 8 correct fifo, 8 skip-ack-loop bag, 8 skip-ack-loop fifo, 9 correct bag, 9 skip-ack-loop fifo, 9 correct fifo, 9 skip-ack-loop bag.

offline

offline: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	400	256	conforms	0.2	4	1 h, 250 GB	0.1	0.1	0.0	merge	brain05, 2026-11-03
3	104976	82944	conforms	8.1	4	1 h, 250 GB	4.4	3.7	0.0	merge	brain05, 2026-11-03
4	95883264	84934656	conforms	2529.4	240	1 h, 250 GB	1326.6	1202.7	0.0	merge	brain05, 2026-11-03

offline: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	400	256	conforms	0.3	4	1 h, 250 GB	0.1	0.2	0.0	merge	brain05, 2026-11-03
3	104976	82944	conforms	8.5	4	1 h, 250 GB	4.7	3.8	0.0	merge	brain05, 2026-11-03

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
4	95883264	84934656	conforms	2605.8	234	1 h, 250 GB	1412.8	1192.9	0.1	merge	brain05, 2026-11-03

offline: pair **skip-ack-loop**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	400	64	16	violation	0.2	4	1 h, 250 GB	0.1	0.1	0.0	merge	brain05, 2026-11-03
3	104976	2304	64	violation	4.6	4	1 h, 250 GB	4.4	0.2	0.0	merge	brain05, 2026-11-03
4	95883264	147456	256	violation	1333.0	240	1 h, 250 GB	1326.6	6.4	0.0	merge	brain05, 2026-11-03

offline: pair **skip-ack-loop**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	400	64	16	violation	0.2	4	1 h, 250 GB	0.1	0.1	0.0	merge	brain05, 2026-11-03
3	104976	2304	64	violation	5.0	4	1 h, 250 GB	4.7	0.2	0.0	merge	brain05, 2026-11-03
4	95883264	147456	256	violation	1419.4	234	1 h, 250 GB	1412.8	6.5	0.0	merge	brain05, 2026-11-03

Not run (offline). planned but never scheduled (grid points beyond the ladder): 4 bag skip-ack-loop, 4 bag correct, 5 bag correct, 5 bag skip-ack-loop, 8 skip-ack-loop bag.

skipped: a smaller size had reached its limit: 5 correct bag, 5 correct fifo, 6 correct bag, 6 correct fifo, 6 skip-ack-loop fifo, 6 skip-ack-loop bag, 7 skip-ack-loop fifo, 7 correct bag, 7 skip-ack-loop bag, 7 correct fifo, 8 correct bag, 8 correct fifo, 8 skip-ack-loop bag, 8 skip-ack-loop fifo, 9 correct bag, 9 skip-ack-loop fifo, 9 correct fifo, 9 skip-ack-loop bag.

refused (an input missing at run time): 5 skip-ack-loop fifo, 5 skip-ack-loop bag.

cross offline, Java implementation against the Rust specification

cross offline, Java implementation against the Rust specification: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	400	256	conforms	5.2	171	10 min, 250 GB	0.1	5.1	0.0	merge	brain01, 2026-11-03

cross offline, Java implementation against the Rust specification: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	400	256	conforms	5.3	178	10 min, 250 GB	0.1	5.1	0.1	merge	brain01, 2026-11-03

cross offline, Java implementation against the Rust specification: pair **skip-ack-loop**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	400	64	16	violation	7.7	151	10 min, 250 GB	0.1	7.5	0.1	merge	brain01, 2026-11-03

cross offline, Java implementation against the Rust specification: pair **skip-ack-loop**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	400	64	16	violation	8.0	163	10 min, 250 GB	0.1	7.7	0.1	merge	brain01, 2026-11-03

Not run (cross offline, Java implementation against the Rust specification). refused (an input missing at run time): 3 fifo correct, 3 fifo skip-ack-loop, 3 bag correct, 3 bag skip-ack-loop.

planned but never scheduled (grid points beyond the ladder): 4 bag skip-ack-loop, 4 bag correct, 5 bag correct, 5 bag skip-ack-loop.

cross offline, Rust implementation against the Java specification

cross offline, Rust implementation against the Java specification: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	400	256	conforms	5.3	167	10 min, 250 GB	5.1	0.1	0.2	merge	brain01, 2026-11-03
3	104976	82944	conforms	159.4	10897	10 min, 250 GB	155.6	3.7	0.1	merge	brain01, 2026-11-03

cross offline, Rust implementation against the Java specification: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	400	256	conforms	5.5	177	10 min, 250 GB	5.3	0.2	0.0	merge	brain01, 2026-11-03
3	104976	82944	conforms	270.5	12238	10 min, 250 GB	266.6	3.8	0.1	merge	brain01, 2026-11-03

cross offline, Rust implementation against the Java specification: pair **skip-ack-loop**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	400	64	16	violation	5.8	167	10 min, 250 GB	5.1	0.1	0.7	merge	brain01, 2026-11-03
3	104976	2304	64	violation	155.9	10897	10 min, 250 GB	155.6	0.2	0.1	merge	brain01, 2026-11-03

cross offline, Rust implementation against the Java specification: pair **skip-ack-loop**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	400	64	16	violation	5.4	177	10 min, 250 GB	5.3	0.1	0.0	merge	brain01, 2026-11-03
3	104976	2304	64	violation	266.9	12238	10 min, 250 GB	266.6	0.2	0.0	merge	brain01, 2026-11-03

Not run (cross offline, Rust implementation against the Java specification). planned but never scheduled (grid points beyond the ladder): 4 bag skip-ack-loop, 4 bag correct, 5 bag correct, 5 bag skip-ack-loop.

cross online, Rust implementation (TraceForge) against the Java specification export

cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	400	256	conforms	0.0	4	1 h, 250 GB		brain01, 2026-11-02
3	104976	82944	conforms	5.1	8	1 h, 250 GB		brain01, 2026-11-02

cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	400	256	conforms	0.0	4	1 h, 250 GB		brain01, 2026-11-02
3	104976	82944	conforms	5.2	6	1 h, 250 GB		brain01, 2026-11-02

cross online, Rust implementation (TraceForge) against the Java specification export: pair **skip-ack-loop**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	400	64	64	violation	0.0	4	1 h, 250 GB	2	brain01, 2026-11-02
3	104976	2304	2304	violation	0.3	4	1 h, 250 GB	3	brain01, 2026-11-02

cross online, Rust implementation (TraceForge) against the Java specification export: pair **skip-ack-loop**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	400	64	64	violation	0.0	4	1 h, 250 GB	2	brain01, 2026-11-02
3	104976	2304	2304	violation	0.3	6	1 h, 250 GB	2	brain01, 2026-11-02

Not run (cross online, Rust implementation (TraceForge) against the Java specification export). planned but never scheduled (grid points beyond the ladder): 4 bag skip-ack-loop, 4 bag correct, 5 bag correct, 5 bag skip-ack-loop.

twins offline, Rust implementation against its Java twin

twins offline, Rust implementation against its Java twin: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	256	256	conforms	5.2	171	10 min, 250 GB	5.1	0.1	0.0	merge	brain01, 2026-11-03
3	82944	82944	conforms	142.8	9253	10 min, 250 GB	139.0	3.7	0.1	merge	brain01, 2026-11-03

twins offline, Rust implementation against its Java twin: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	256	256	conforms	5.3	178	10 min, 250 GB	5.1	0.2	0.0	merge	brain01, 2026-11-03
3	82944	82944	conforms	269.7	10018	10 min, 250 GB	265.9	3.8	0.0	merge	brain01, 2026-11-03

twins offline, Rust implementation against its Java twin: pair **skip-ack-loop**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	64	64	conforms	7.8	151	10 min, 250 GB	7.5	0.1	0.2	merge	brain01, 2026-11-03
3	2304	2304	conforms	19.0	2187	10 min, 250 GB	18.7	0.2	0.0	merge	brain01, 2026-11-03

twins offline, Rust implementation against its Java twin: pair **skip-ack-loop**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	64	64	conforms	7.8	163	10 min, 250 GB	7.7	0.1	0.0	merge	brain01, 2026-11-03
3	2304	2304	conforms	10.3	572	10 min, 250 GB	10.0	0.2	0.0	merge	brain01, 2026-11-03

twins offline, Rust implementation against its Java twin: pair **skip-ack-loop**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	256	64	16	violation	5.2	171	10 min, 250 GB	5.1	0.1	0.0	merge	brain01, 2026-11-03
3	82944	2304	64	violation	139.4	9253	10 min, 250 GB	139.0	0.2	0.2	merge	brain01, 2026-11-03

twins offline, Rust implementation against its Java twin: pair **skip-ack-loop**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	256	64	16	violation	5.3	178	10 min, 250 GB	5.1	0.1	0.1	merge	brain01, 2026-11-03
3	82944	2304	64	violation	266.4	10018	10 min, 250 GB	265.9	0.2	0.3	merge	brain01, 2026-11-03

Not run (twins offline, Rust implementation against its Java twin). planned but never scheduled (grid points beyond the ladder): 4 bag skip-ack-loop, 4 bag correct.

twins offline, Java implementation against its Rust twin

twins offline, Java implementation against its Rust twin: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	256	256	conforms	5.2	171	10 min, 250 GB	0.1	5.1	0.0	merge	brain01, 2026-11-03
3	82944	82944	conforms	142.9	9253	10 min, 250 GB	3.7	139.0	0.2	merge	brain01, 2026-11-03

twins offline, Java implementation against its Rust twin: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	256	256	conforms	5.3	178	10 min, 250 GB	0.2	5.1	0.0	merge	brain01, 2026-11-03
3	82944	82944	conforms	270.3	10018	10 min, 250 GB	3.8	265.9	0.6	merge	brain01, 2026-11-03

twins offline, Java implementation against its Rust twin: pair **skip-ack-loop**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	64	64	conforms	7.6	151	10 min, 250 GB	0.1	7.5	0.0	merge	brain01, 2026-11-03
3	2304	2304	conforms	19.5	2187	10 min, 250 GB	0.2	18.7	0.5	merge	brain01, 2026-11-03

twins offline, Java implementation against its Rust twin: pair **skip-ack-loop**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	64	64	conforms	7.8	163	10 min, 250 GB	0.1	7.7	0.0	merge	brain01, 2026-11-03
3	2304	2304	conforms	10.4	572	10 min, 250 GB	0.2	10.0	0.2	merge	brain01, 2026-11-03

twins offline, Java implementation against its Rust twin: pair **skip-ack-loop**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	256	64	16	violation	7.6	151	10 min, 250 GB	0.1	7.5	0.0	merge	brain01, 2026-11-03
3	82944	2304	64	violation	22.6	2187	10 min, 250 GB	3.7	18.7	0.1	merge	brain01, 2026-11-03

twins offline, Java implementation against its Rust twin: pair skip-ack-loop, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	256	64	16	violation	7.9	163	10 min, 250 GB	0.2	7.7	0.0	merge	brain01, 2026-11-03
3	82944	2304	64	violation	13.8	572	10 min, 250 GB	3.8	10.0	0.1	merge	brain01, 2026-11-03

Not run (twins offline, Java implementation against its Rust twin). planned but never scheduled (grid points beyond the ladder): 4 bag skip-ack-loop, 4 bag correct.

context-free twins offline, Rust implementation against the context-free Java port

context-free twins offline, Rust implementation against the context-free Java port: pair correct, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	256	256	conforms	7.2	158	10 min, 250 GB	7.0	0.1	0.1	merge	brain01, 2026-11-03
3	82944	82944	conforms	146.3	9068	10 min, 250 GB	142.5	3.7	0.1	merge	brain01, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair correct, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	256	256	conforms	6.3	176	10 min, 250 GB	6.2	0.2	0.0	merge	brain01, 2026-11-03
3	82944	82944	conforms	164.8	9819	10 min, 250 GB	160.8	3.8	0.3	merge	brain01, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair skip-ack-loop, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	64	64	conforms	4.4	156	10 min, 250 GB	4.3	0.1	0.0	merge	brain01, 2026-11-03
3	2304	2304	conforms	11.9	2944	10 min, 250 GB	11.7	0.2	0.0	merge	brain01, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair skip-ack-loop, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	64	64	conforms	7.3	156	10 min, 250 GB	7.1	0.1	0.1	merge	brain01, 2026-11-03
3	2304	2304	conforms	12.4	2889	10 min, 250 GB	12.1	0.2	0.1	merge	brain01, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair skip-ack-loop, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	256	64	16	violation	7.3	158	10 min, 250 GB	7.0	0.1	0.3	merge	brain01, 2026-11-03
3	82944	2304	64	violation	142.8	9068	10 min, 250 GB	142.5	0.2	0.0	merge	brain01, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair skip-ack-loop, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	256	64	16	violation	6.3	176	10 min, 250 GB	6.2	0.1	0.0	merge	brain01, 2026-11-03
3	82944	2304	64	violation	161.0	9819	10 min, 250 GB	160.8	0.2	0.0	merge	brain01, 2026-11-03

Not run (context-free twins offline, Rust implementation against the context-free Java port). planned but never scheduled (grid points beyond the ladder): 4 bag skip-ack-loop, 4 bag correct.

context-free twins offline, Java port against the Rust implementation

context-free twins offline, Java port against the Rust implementation: pair correct, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	256	256	conforms	7.0	158	10 min, 250 GB	0.1	7.0	0.0	merge	brain01, 2026-11-03
3	82944	82944	conforms	146.3	9068	10 min, 250 GB	3.7	142.5	0.0	merge	brain01, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair correct, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	256	256	conforms	6.4	176	10 min, 250 GB	0.2	6.2	0.0	merge	brain01, 2026-11-03
3	82944	82944	conforms	164.6	9819	10 min, 250 GB	3.8	160.8	0.1	merge	brain01, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair skip-ack-loop, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	64	64	conforms	4.5	156	10 min, 250 GB	0.1	4.3	0.2	merge	brain01, 2026-11-03
3	2304	2304	conforms	11.9	2944	10 min, 250 GB	0.2	11.7	0.0	merge	brain01, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair skip-ack-loop, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	64	64	conforms	7.2	156	10 min, 250 GB	0.1	7.1	0.0	merge	brain01, 2026-11-03
3	2304	2304	conforms	12.4	2889	10 min, 250 GB	0.2	12.1	0.0	merge	brain01, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair skip-ack-loop, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	256	64	16	violation	4.5	156	10 min, 250 GB	0.1	4.3	0.1	merge	brain01, 2026-11-03
3	82944	2304	64	violation	15.4	2944	10 min, 250 GB	3.7	11.7	0.0	merge	brain01, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair skip-ack-loop, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	256	64	16	violation	7.3	156	10 min, 250 GB	0.2	7.1	0.0	merge	brain01, 2026-11-03
3	82944	2304	64	violation	15.9	2889	10 min, 250 GB	3.8	12.1	0.0	merge	brain01, 2026-11-03

Not run (context-free twins offline, Java port against the Rust implementation). planned but never scheduled (grid points beyond the ladder): 4 bag skip-ack-loop, 4 bag correct.

context-free twins online, Java port (JMC) against the Rust implementation's export

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	256	256	conforms	8.6	174	1 h, 250 GB		brain01, 2026-11-03
3	82944	82944	conforms	188.6	10872	1 h, 250 GB		brain01, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	256	256	conforms	8.2	168	1 h, 250 GB		brain01, 2026-11-03
3	82944	82944	conforms	207.3	11897	1 h, 250 GB		brain01, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **skip-ack-loop**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	64	64	conforms	8.5	166	1 h, 250 GB		brain01, 2026-11-03
3	2304	2304	conforms	16.4	2602	1 h, 250 GB		brain01, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **skip-ack-loop**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	64	64	conforms	8.0	171	1 h, 250 GB		brain01, 2026-11-03
3	2304	2304	conforms	17.3	2482	1 h, 250 GB		brain01, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **skip-ack-loop**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	256	64	64	violation	7.4	178	1 h, 250 GB	192	brain01, 2026-11-03
3	82944	2304	2304	violation	17.7	2491	1 h, 250 GB	312	brain01, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **skip-ack-loop**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	256	64	64	violation	7.6	165	1 h, 250 GB	168	brain01, 2026-11-03
3	82944	2304	2304	violation	18.0	2445	1 h, 250 GB	199	brain01, 2026-11-03

Not run (context-free twins online, Java port (JMC) against the Rust implementation's export). planned but never scheduled (grid points beyond the ladder): 4 bag skip-ack-loop, 4 bag correct.

context-free twins online, Rust implementation (TraceForge) against the Java port's export

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	256	256	conforms	0.0	4	1 h, 250 GB		brain01, 2026-11-02
3	82944	82944	conforms	5.2	4	1 h, 250 GB		brain01, 2026-11-02

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	256	256	conforms	0.0	4	1 h, 250 GB		brain01, 2026-11-02
3	82944	82944	conforms	5.1	4	1 h, 250 GB		brain01, 2026-11-02

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **skip-ack-loop**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	64	64	conforms	0.0	4	1 h, 250 GB		brain01, 2026-11-02
3	2304	2304	conforms	0.3	4	1 h, 250 GB		brain01, 2026-11-02

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **skip-ack-loop**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	64	64	conforms	0.0	4	1 h, 250 GB		brain01, 2026-11-02
3	2304	2304	conforms	0.3	4	1 h, 250 GB		brain01, 2026-11-02

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **skip-ack-loop**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	256	64	64	violation	0.0	4	1 h, 250 GB	2	brain01, 2026-11-02
3	82944	2304	2304	violation	0.3	4	1 h, 250 GB	2	brain01, 2026-11-02

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **skip-ack-loop**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	256	64	64	violation	0.0	4	1 h, 250 GB	2	brain01, 2026-11-02
3	82944	2304	2304	violation	0.3	4	1 h, 250 GB	2	brain01, 2026-11-02

Not run (context-free twins online, Rust implementation (TraceForge) against the Java port's export). planned but never scheduled (grid points beyond the ladder): 4 bag skip-ack-loop, 4 bag correct.